Week 3: Linux Logs, Sysmon, auditd and Snort Setup

Theory Notes, Developing Basic Understanding in SOC

Question 1. What is Endpoint Detection and Response (EDR)? Why is endpoint data critical for a SOC analyst? How does EDR differ from traditional antivirus?

What is EDR

EDR is a security tool that is installed on endpoints such as laptops, desktops and servers. It quietly watches what happens on the machine all the time, and it helps the security team find a problem and deal with it.

- It records things like which programs were started, which files were changed, which network connections were made and who logged in.
- It looks at this activity for anything suspicious and raises an alert when it finds something.
- If there is a real threat, the analyst can act from the same place, for example by isolating the machine or stopping the process.
- Some well known examples are CrowdStrike Falcon, Microsoft Defender for Endpoint and SentinelOne.

What EDR does

Function	What it means
Monitoring	Keeps recording activity on the endpoint all the time, not only when a scan runs
Detection	Finds suspicious behaviour using rules, known indicators and analysis
Investigation	Shows a timeline of what happened, so I can follow the attack from the start
Response	Lets the analyst isolate the machine, stop a process or remove a file
Threat hunting	Lets analysts search the stored data to look for activity that was missed

Why endpoint data matters to a SOC analyst

Most attacks end up running on a computer at some point, so the endpoint is where the most useful evidence is found. Network logs can show that two machines talked to each other, but they cannot tell me which program did it or what it did on the machine. Endpoint data fills that gap.

- It shows the exact process, command and user behind an activity, which helps me understand what really happened.
- It helps me put the attack in order, for example which program started first, what it connected to and what file it created.
- It helps me notice signs of an attacker staying in the system or moving to other machines.
- It helps me find out how far the problem has spread by checking if other machines show the same activity.
- It is also a useful record for forensic work later.

EDR compared with traditional antivirus

Point	Traditional antivirus	EDR
Main aim	Stop known malware	Detect, investigate and respond to threats
How it detects	Mostly by matching signatures and file hashes	By watching behaviour, along with indicators and analysis
New or unknown threats	Often misses them	Has a better chance, because it looks at behaviour
Visibility	Limited, mostly scan results	Detailed record of processes, files, network and registry
Investigation	Very little to work with	Gives a full timeline and process tree
Response	Quarantine or delete the file	Isolate the machine, stop the process, and more
Mostly used by	Everyday users and basic IT	SOC and incident response teams

To put it simply, antivirus tells me that a file was blocked, while EDR helps me understand the whole story around it. Many modern products now include both.